

AI AUTOMATION PROJECT

TravelOS

An AI-powered Telegram bot that plans your entire trip in
seconds

Project: TravelOS Trip Planner Bot

Stack: n8n · Google Gemini 2.5 · Google Places API · Telegram

Author: Noel George Mathew

GitHub: github.com/noe7george

Project Overview

TravelOS is a conversational AI travel assistant that lives inside Telegram. Users describe their trip in natural language — the bot extracts structured details, fetches real attractions from Google Places, generates a personalized day-wise itinerary with Gemini, and stores everything in Google Sheets for later recall.

Problem Statement

Planning a trip today means juggling 5–10 browser tabs for weather, attractions, restaurants, hotels, and itineraries. TravelOS collapses that entire workflow into a single chat interface — from "I want to go to Munnar" to a fully-scheduled day-wise plan in under 30 seconds.

Key Features

Natural Language Parsing

Extracts trip details from free-text messages using Gemini

Smart Field Detection

Detects missing fields and asks the user for them

Attraction Discovery

Fetches top-rated places from Google Places API

Photo Carousel

Sends 5 real attraction photos in a single media group

Day-Wise Itinerary

Full schedule with times, costs, and activities

Google Sheets Storage

Persists trips and activities for later recall

Impact: What used to take hours of manual research now takes a single Telegram message.

System Architecture

The workflow runs entirely on n8n — a visual automation platform. Every message triggers a chain of nodes that parse, enrich, generate, and store trip data.

User Flow

- 1 User sends a message on Telegram**
"Plan a trip from Kochi to Munnar by car for 2 people with ₹100000"
- 2 Switch node routes by intent**
Detects whether the user wants a new trip, to view an itinerary, or a day-wise view
- 3 Gemini extracts structured data**
Converts the free text into JSON: origin, destination, dates, travelers, budget, style
- 4 Completeness check**
If any required field is missing, the bot asks for it and re-parses on the next reply
- 5 Google Places API call**
Fetches top-rated attractions with photos, ratings, and addresses
- 6 Gemini generates a welcome summary**
Writes a themed intro mentioning 4 attractions in a friendly tone
- 7 Telegram media group**
Sends 5 attraction photos with a combined caption listing names and ratings
- 8 Gemini creates the full itinerary**
Day-by-day schedule with activities, meals, and rest breaks
- 9 Google Sheets persistence**
Saves the trip record and each activity as a separate row
- 10 Confirmation message**
Summarizes the trip and offers follow-up options

Technology Stack

Every component was chosen for speed of development, reliability, and cost efficiency.

Layer	Technology	Purpose
Orchestration	n8n	Visual workflow automation
LLM	Google Gemini 2.5 Flash	NL parsing & itinerary generation
Attractions	Google Places API (New)	Real-time attraction discovery
Storage	Google Sheets	Persistent trip + activity records
Interface	Telegram Bot API	Chat-based user interaction

Notable Engineering Decisions

- ▶ **JSON-only LLM output** — Gemini is instructed to return only valid JSON, never markdown
- ▶ **Completeness gating** — The workflow refuses to generate itineraries until required fields exist
- ▶ **Currency defaulting** — If the user mentions a budget without a currency, INR is assumed
- ▶ **Review-count sorting** — Attractions are ranked by total reviews first, rating second
- ▶ **Media group API** — 5 photos sent in one Telegram message instead of 5 separate ones

Why n8n? It allowed testing the entire flow — API calls, LLM prompts, and conditional logic — visually, without writing a backend. The result: a working product in a fraction of the time.

Demo Walkthrough

1. Welcome Message

When a user sends /start or hi, the bot replies with a friendly welcome and a fill-in-the-blank template. The user simply copies, fills in their details, and sends it back.

2. Natural Conversation

Users don't need to learn any format. Example input:

Origin: Kottayam · Destination: Kumarakom · Transport Mode: Flight
Start Date: 1/8/2026 · End Date: 6/8/2026
Travelers: 2 · Budget: 30000

TravelOS parses all fields correctly, even from loosely formatted text.

3. AI-Generated Destination Preview

Before the itinerary, the bot sends:

- ▶ A 2-sentence themed intro to the destination
- ▶ 4 curated attractions with one-line descriptions
- ▶ Top hotel and restaurant recommendations
- ▶ 5 real photos from Google Places in a swipeable media group

4. Full Itinerary

Gemini generates a complete day-by-day schedule with:

- ▶ Timestamps for each activity
- ▶ Natural-language descriptions (not just labels)
- ▶ Meals, rest breaks, and sightseeing balanced throughout
- ▶ Budget-aware cost estimates

Skills Demonstrated

Technical

- ▶ **LLM Orchestration** — Chaining two Gemini calls with different system prompts (extraction vs generation)
- ▶ **Structured Output Parsing** — Enforcing JSON schema from an LLM and safely handling markdown-wrapped responses
- ▶ **Conditional Workflows** — Branching by intent and by data completeness
- ▶ **REST API Integration** — Google Places, Telegram Bot, Google Sheets
- ▶ **Conversational UX Design** — Intent routing, graceful fallbacks, "home" and "help" commands
- ▶ **Data Persistence** — Multi-table writes with foreign key relationships

Product

- ▶ **Problem framing** — Turning a fuzzy real-world need into a testable product
- ▶ **Prompt engineering** — Extensive system prompts with examples, edge cases, and validation rules
- ▶ **API cost management** — Preview limits, review-count sorting, caching via Sheets
- ▶ **User empathy** — Copy-paste templates, swipe navigation, clear follow-up options

What Could Come Next

- ▶ Weather API integration for daily forecasts
- ▶ Multi-language itineraries
- ▶ PDF export of the itinerary
- ▶ Hotel & restaurant suggestions beyond attractions
- ▶ Voice input via Whisper
- ▶ Currency conversion for international trips

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Thank You

Questions, feedback, or collaboration?

Noel George Mathew

github.com/noe7george

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